STATISTICAL BUSINESS ANALYSIS

Customer Churn Analysis

Executive Summary

The dataset contains 500 customers, of whom 53 (10.6%) are marked as churned. Average tenure is 36.5 months and average monthly charges are 113.64. The churn proportion has a 95% Wilson confidence interval of 8.2% to 13.6%. Group tests assess whether charges and tenure differ by churn status and whether churn is associated with contract/payment categories. The findings are associations within this sample, not proof that a variable causes churn.

1. Project Overview and Scope

Objective: quantify customer churn, describe the customer and billing profile, test differences between churned and retained customers, measure linear associations, estimate confidence intervals, and build regression models. **Scope adjustment:** the supplied file contains no sales, marketing-spend or revenue variables. Accordingly, those requested analyses cannot be performed without another dataset; this report analyzes churn, tenure and charges instead.

2. Data Overview and Preparation

Rows: 500; columns: 9; missing cells: 0. CustomerID is an identifier, not a numeric predictor. Churn is binary (1=yes, 0=no); SeniorCitizen is a binary indicator. No rows were removed. A derived ChurnLabel text field was added to business_data.csv for readability.

3. Descriptive Statistics

Variable	Mean	Median	Mode	Std. dev.	Min	Max
Tenure	36.53	37.00	3	20.67	1	71
MonthlyCharges	113.64	115.00	115	51.80	20	199
TotalCharges	4237.88	4182.50	4023	2260.62	159	7992
SeniorCitizen	0.50	0.00	0	0.50	0	1
Churn	0.11	0.00	0	0.31	0	1

Categorical modes: Contract=One year; PaymentMethod=Credit Card; PaperlessBilling=No.

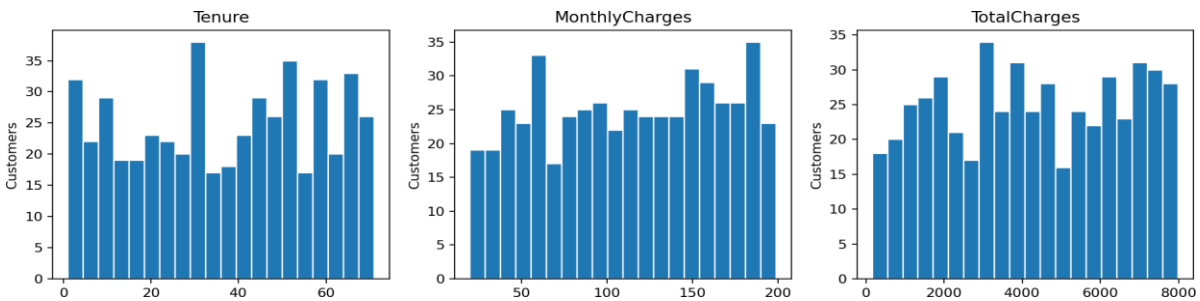
4. Distribution and Normality

Tenure: Shapiro-Wilk W=0.9496, p=5.1026e-12; evidence against normality at 5%.

MonthlyCharges: Shapiro-Wilk W=0.9520, p=1.1492e-11; evidence against normality at 5%.

TotalCharges: Shapiro-Wilk W=0.9514, p=9.383e-12; evidence against normality at 5%.

Histograms show the observed distributions. Normality tests are sensitive to sample size; Welch t-tests are used for group mean comparisons because they do not assume equal variances.



5. Correlation Analysis

Pearson correlation measures linear association from -1 to +1. It does not establish causation. Key correlations:

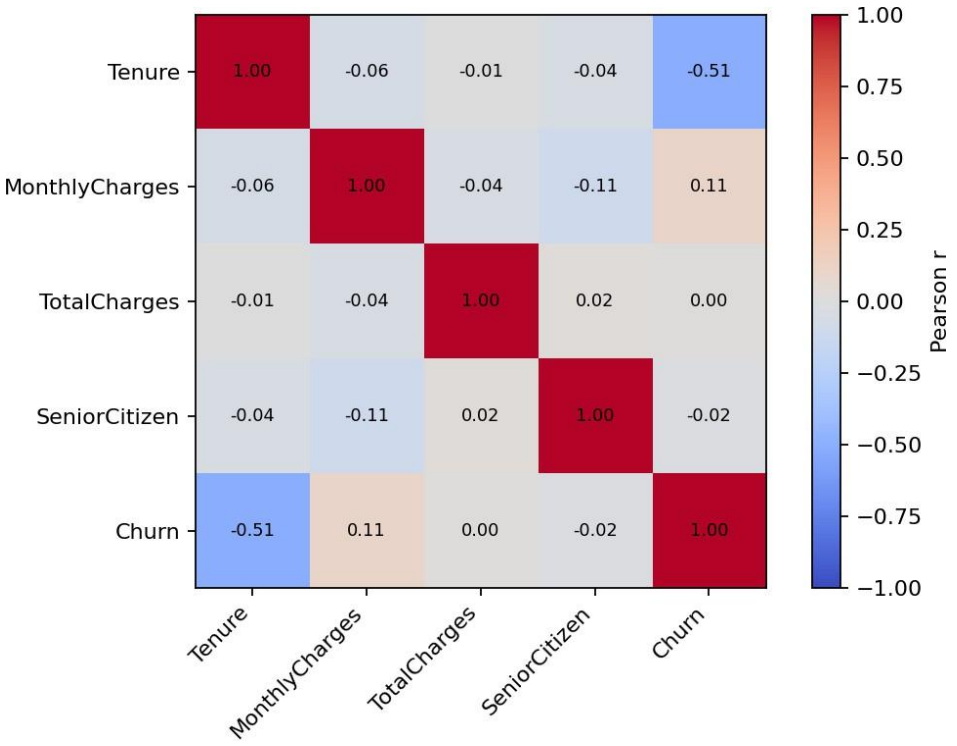
Tenure vs TotalCharges: r = -0.006.

MonthlyCharges vs TotalCharges: r = -0.042.

Tenure vs MonthlyCharges: r = -0.060.

Churn vs Tenure: r = -0.509.

Churn vs MonthlyCharges: r = 0.107.



6. Hypothesis Testing

Tests use alpha=0.05. Welch independent-samples t-tests compare numeric means by churn status; one-way ANOVA compares mean monthly charges across contract groups; chi-square tests assess independence of categorical variables and churn.

Welch t-test: Tenure (Churn=1 vs 0): statistic=-34.535, p=8.9823e-130 — significant. Mean churned=6.00; retained=40.15

Difference (churned minus retained)=-34.15; 95% CI [-36.10, -32.21].

Welch t-test: MonthlyCharges (Churn=1 vs 0): statistic=2.646, p=0.010079 — significant. Mean churned=129.77; retained=111.72

Difference (churned minus retained)=18.05; 95% CI [4.44, 31.66].

Welch t-test: TotalCharges (Churn=1 vs 0): statistic=0.095, p=0.92497 — not significant. Mean churned=4265.75; retained=4234.58

Difference (churned minus retained)=31.18; 95% CI [-627.48, 689.84].

One-way ANOVA: MonthlyCharges by Contract: statistic=0.031, p=0.96921 — not significant. Month-to-month: 112.83; One year: 114.01; Two year: 114.11

Chi-square: Contract vs Churn: statistic=27.715, p=9.5874e-07 — significant. chi2=27.715, df=2; contingency table rows=contract, cols=churn

Chi-square: PaymentMethod vs Churn: statistic=3.870, p=0.14446 — not significant. chi2=3.870, df=2

7. Confidence Intervals

Tenure: mean=36.53; 95% CI [34.72, 38.35]; margin of error=±1.82.

MonthlyCharges: mean=113.64; 95% CI [109.08, 118.19]; margin of error=±4.55.

TotalCharges: mean=4237.88; 95% CI [4039.25, 4436.51]; margin of error=±198.63.

Churn rate: 10.6%; 95% Wilson CI [8.2%, 13.6%]. Wilson interval is used for a binomial proportion.

8. Regression Analysis

Multiple linear regression: TotalCharges ~ Tenure + MonthlyCharges. R-squared=0.002; adjusted R-squared=-0.002; model F-test p=0.6304.

Intercept: coefficient=4482.871, p=1.3617e-39, 95% CI [3871.708, 5094.035].

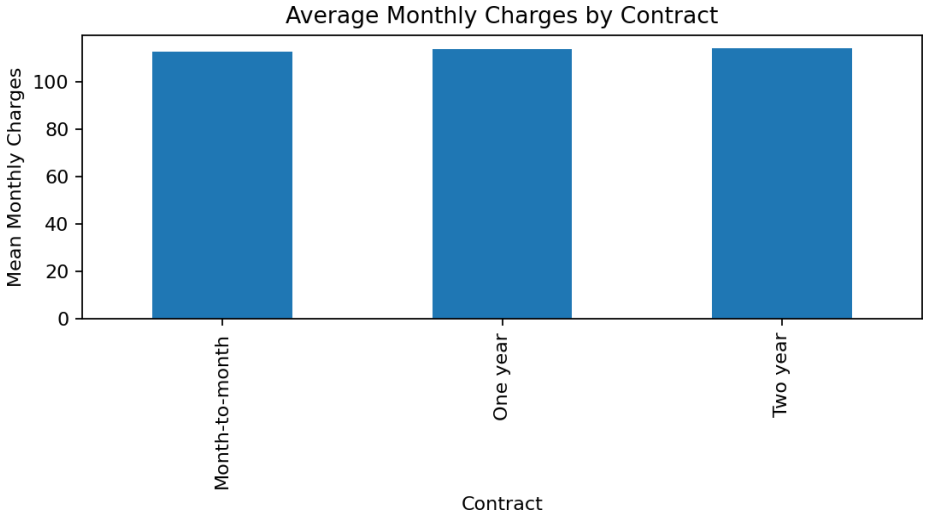
Tenure: coefficient=-0.900, p=0.85465, 95% CI [-10.548, 8.748].

MonthlyCharges: coefficient=-1.867, p=0.34121, 95% CI [-5.716, 1.983].

Supplementary churn classifier: logistic regression on a stratified 75/25 train/test split. Holdout accuracy=0.944; ROC-AUC=0.987. This is a baseline model on a small sample; performance should be validated on a larger, temporally separated dataset before operational use.

9. Business Insights and Recommendations

- Prioritize retention analysis around variables that show statistically significant group differences or associations in the hypothesis-test results.
- Use tenure and billing signals to define customer cohorts; compare churn rates by contract and payment method before designing interventions.
- If a contract/payment category is statistically associated with churn, investigate customer experience and offer design within that segment rather than assuming the category causes churn.
- Track churn rate and its confidence interval monthly, alongside cohort-level tenure and charge metrics.
- Validate any predictive model on more records and out-of-time data; monitor recall, precision and calibration, not accuracy alone.



10. Methodology, Limitations and Validation

Methods: descriptive statistics; histograms; Shapiro-Wilk normality checks; Pearson correlation; Welch t-tests; one-way ANOVA; chi-square tests; 95% t-based confidence intervals for means; Wilson confidence interval for churn proportion; OLS linear regression; and a supplementary logistic classifier. Limitations: cross-sectional sample of 500; no time variable; no sales, marketing expenditure, revenue, customer acquisition cost intervention data; observational associations cannot establish causal impact; no external validation sample. The OLS model explains charge variation, not churn causation.

11. Setup Instructions and Code Structure

1. Install Python 3.10+ and create a virtual environment.
2. Run `pip install -r requirements.txt`.
3. Keep customer_churn.csv in the same directory as statistical_analysis.ipynb.
4. Launch Jupyter with `jupyter notebook`.
5. Run all cells from top to bottom.

6. Compare notebook output with hypothesis_tests_results.txt and this report.

Files: statistical_analysis.ipynb (reproducible analysis); business_data.csv (analysis-ready copy); statistical_report.pdf (narrative findings); hypothesis_tests_results.txt (test details); requirements.txt (dependencies).

12. Testing Evidence

Validation checks: expected dataset shape=(500, 9); missing-value count=0; Churn values=[0, 1]; total churn counts=53+447=500; confidence interval lower bounds are below upper bounds; the train/test split is stratified. Re-run the notebook to reproduce numeric results. Screenshots are not embedded because notebook execution screenshots are environment-specific; the histogram, correlation heatmap and contract comparison charts provide visual evidence in this report.

Conclusion

This dataset supports a focused statistical review of customer churn and billing behavior. The most actionable next step is to connect churn patterns to customer journeys and test targeted retention actions with controlled measurement. To analyze marketing effectiveness or sales/revenue impact, provide a dataset containing customer-level or period-level marketing spend and revenue/sales measures.